

习题 1 第 3, 4, 9, 10, 14, 16

3.

(1)

$$t, -t, 1, -1, 0$$

$$R_1(s, t) = s t E(A^2 X) + (s+t) E[(A-A^2)X] + 1 - 2E(A) + E(A^2)$$

(2)

$$P(X(1)=1) = P(A+0 \cdot B=1)$$

$$E(A) = \frac{1}{2} \quad E(A^2) = \frac{1}{2}$$

$$E(A^2 X) = \frac{1}{2}$$

$$E(X^2) = 1 + 1 = 2$$

$$E(A-A^2) = 0$$

$$R_1(s, t) = s t \cdot \frac{1}{2} \cdot 2 + 0 + 1 - 1 + \frac{1}{2} = s t + \frac{1}{2}$$

P	A	B	X(1)
$\frac{1}{3}$	1	X	1
$\frac{1}{3}$	-1	X	-1
$\frac{1}{3}$	0	1	1
$\frac{1}{3}$	0	-1	-1
$\frac{1}{3}$	0	0	0

$$P(X(1)=1) = \frac{1}{3} + \frac{1}{3} = \frac{2}{3}$$

$$P(X(2)=1) = P(A=0, B=1) = \frac{1}{3}$$

$$P(X(1)=1, X(2)=1) = P(A=0, B=1) = \frac{1}{3}$$

4.

(1)

$$Z(1) = AX + 1 - A \quad Z(2) = 2AX + 1 - A$$

$$P(Z(1) < 1) = P(A=0) \cdot 0 + P(A=1) P(X < 1)$$

$$= \frac{1}{2} P(X < 1)$$

$$= \frac{1}{2} \times \frac{1}{2}$$

$$= 0.25$$

$$P(Z(2) < 2) = P(A=0) + P(A=1) P(X < 1)$$

$$= 0.5 + 0.25$$

$$= 0.75$$

$$P(Z(1) < 1, Z(2) < 2) = P(A=0) P(X < 1) = 0.25$$

$$(2) \mu_Z(t) = t E(AX) + 1 - E(A)$$

$$E(A) = \frac{1}{2}$$

$$E(AX) = \frac{1}{2} \times 1 = \frac{1}{2}$$

$$\mu_Z(t) = \frac{1}{2} t + \frac{1}{2}$$

9.

$$\mu_X(t) = t E(A) + E(B)$$

$$= \mu t + \mu$$

$$R_X(s, t) = s t E(A^2) + (s+t) E(AB) + E(B^2)$$

$$E(A) = \mu^2 + \sigma^2$$

$$E(A^2) = \mu^2$$

$$R_X(s, t) = s t (\mu^2 + \sigma^2) + (s+t) \mu^2 + \mu^2 + \sigma^2$$

$$= (s+t) (\mu^2 + \sigma^2) + (s+t) \mu^2$$

$$C_X(s, t) = R_X(s, t) - \mu_X(s) \mu_X(t)$$

$$= (s+t) \sigma^2 + (s+t) (\mu^2 + \sigma^2) - (\mu^2 + \sigma^2) (\mu^2 + \sigma^2)$$

$$= (s+t) \sigma^2$$

(2)

$X(t)$ 是正态分布 A, B 线性组合, 所以是正态过程

$$\mu_X(t) = E[X(t)] = 0 \quad \mu_X(t) = t^2 + 1$$

$$R_X(t, s) = E[X(t)X(s)] = 0 \quad X(t) \sim N(0, t^2 + 1)$$

$$X(t) - X(s) = A(t-s) \quad E[X(t) - X(s)] = 0 \quad D[A(t-s)] = (t-s)^2$$

$$X(t) + X(s) = A(t+s) + B \quad E[X(t) + X(s)] = 0$$

$$D[X(t) + X(s)] = (t+s)^2 D(A) + D(B)$$

$$= 4 + (t+s)^2$$

$$X(t) \sim N(0, t^2 + 1)$$

$$X(t) - X(s) \sim N(0, t^2 - s^2)$$

$$X(t) + X(s) \sim N(0, t^2 + s^2 + 4)$$

10.

(1)

$$P(Y_n=y) = C_3^y p^y (1-p)^{3-y} \quad y=0,1,2,3$$

(2)

$$Y_1 = X_1 + X_2 + X_3$$

$$Y_0 = X_0 = 2$$

Y_1 与 Y_0 无关

$$P(Y_1=y | Y_0=2) = C_3^y p^y (1-p)^{3-y} \quad y=0,1,2,3$$

$$Y_0 = X_0 + X_1 + X_2$$

$$Y_1 = X_1 + X_2 + X_3$$

$$P(Y_0=2) = C_3^2 p^2 (1-p)^1 = 3p^2(1-p)$$

$$P(Y_1=0, Y_0=2) = 0$$

$$\begin{aligned} P(Y_1=1, Y_0=2) &= [P(X_1=0, X_2=1, X_3=0) \\ &\quad + P(X_1=1, X_2=0, X_3=0)] P(X_0=1) \\ &= 2p^2(1-p)^2 \end{aligned}$$

$$P(Y_1=2, Y_0=2) = 2p^3(1-p) + p^2(1-p)^2$$

$$\begin{aligned} P(Y_1=3, Y_0=2) &= P(X_0=0, X_1=1, X_2=1, X_3=1) \\ &= p^3(1-p) \end{aligned}$$

~~P(Y_1=0, Y_0=2)~~

$Y_1 Y_0=2$	0	1	2	3
P	0	$\frac{2}{3}(1-p)$	$\frac{1+p}{3}$	$\frac{p}{3}$

(3)

$$P(Y_0=1, Y_1=0, Y_2=1)$$

$$= P(X_0=1, X_1=0, X_2=0, X_3=0, X_4=1)$$

$$= p^2(1-p)^3$$

14

$$E(Y_n) = \sum_{y=0}^3 y C_3^y p^y (1-p)^{3-y} = 3p$$

$$C_Y(m, n) = E(Y_m Y_n) - E(Y_m) E(Y_n)$$

$$= \begin{cases} (3-|n-m|)p(1-p) & |n-m| < 3 \\ 0 & |n-m| \geq 3 \end{cases}$$

14.

$$\mu_z(t) = a(t)\mu_x(t) + b(t)\mu_y(t) + c(t)$$

$$R_z(s, t) = E[z(s)z(t)]$$

$$= a(s)a(t)E[X(s)X(t)] + a(s)b(t)\mu_x(s)\mu_y(t) + a(s)c(t)\mu_x(s)$$

$$+ a(t)b(s)\mu_x(t)\mu_y(s) + b(s)b(t)[C_Y(s, t) + \mu_y(s)\mu_y(t)]$$

$$+ b(s)c(t)\mu_y(s) + a(t)c(s)\mu_x(t)$$

$$+ b(t)c(s)\mu_y(t) + c(s)c(t)$$

$$C_z(s, t) = R_z(s, t) - \mu_z(s)\mu_z(t)$$

$$= a(s)a(t)C_X(s, t) + b(s)b(t)C_Y(s, t)$$

16.

$$\mu_z(t) = E[X(t)]E[Y(t)]$$

$$= \mu_X(t)\mu_Y(t)$$

$$R_z(s, t) = E[z(s)z(t)]$$

$$= R_X(s, t)R_Y(s, t)$$

$$R_{XZ}(s, t) = E[X(s)z(t)]$$

$$= E[X(s)X(t)Y(t)]$$

$$= R_X(s, t)\mu_Y(t)$$